Minseok Kim

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Education

Ajou University *Mar 2021 – Feb 2027*
B.S. in Software and Computer Engineering (expected)
GPA: **4.25/4.50** · Major GPA: **4.22/4.50**

Research Interests

Edge AI System, LLM Inference Optimization, Vision-Language Models, On-Device AI, GPU Computing, Algorithm

Skills

C/C++, Python, Swift, SQL, PyTorch

Experience

iKnow Lab., Ajou University *Jul 2025 – Mar 2026*
Undergraduate Researcher · Advisor: Prof. Hyunsouk Cho

- Designed controlled experiments and synthetic-data generation pipelines to study spatial reasoning in vision-language models.
- Implemented multi-model inference and attention analysis, and participated in regular lab seminars.

AI·SW Maestro, 17th Cohort (Seoul) *Apr 2026 – Present*
Trainee · On-Device AI Development

Hosted by the Ministry of Science and ICT (MSIT). Organized by the Institute of Information & Communications Technology Planning & Evaluation (IITP) and The Federation of Korean Information Industries (FKII).

- Developing Byeolmuri, an on-device AI character app for iOS, in a team-based training program, with responsibility for AI development, on-device evaluation, and improvements informed by TestFlight user feedback.

Republic of Korea Army *Mar 2022 – Sep 2023*
CERT Soldier · Cyber Network Operations Center (CNOC)

- Monitored network access control, blocked unauthorized devices, and handled security events at the Armed Forces Command and Communications Command.
- Completed mandatory service as a Sergeant.

Publication

Diagnosing Axis-Dependent Spatial Bias in Vision-Language Models
Minseok Kim, Hyunsouk Cho
Korea Computer Congress (KCC 2026), Jeju, South Korea
Undergraduate Paper Competition, Poster Presentation · Jun 2026

- Observed an axis-dependent accuracy gap of up to **52.12 percentage points** across four vision-language models.
- Using **2,000 controlled samples**, found a negative correlation between patch sequence distance and inter-object attention (**Spearman's $\rho = -0.782$**); accuracy partially rebounded at the longest distance.

Activities and Projects

Byeolmuri — On-Device AI Character App *May 2026 – Present*

AI Developer · AI · SW Maestro Team Project

- Built an iOS on-device agent with Swift and LiteRT-LM, SQLite-backed memory, and validated tool execution.
- Reduced false tool activation from **45.83% to 4.17%** through intent classification and embedding-based tool selection, maintaining approximately **94%** intent classification performance on the evaluation set.
- Reduced mean time to first token by **74.1% (4.920 s to 1.274 s)** through fixed-prompt KV cache reuse on iPhone 17 CPU across 30 requests, with matching outputs in all 15 response pairs.
- Integrated device memory measurements into the evaluation harness and added streaming responses following TestFlight feedback.

TOWA — AI Manga Translation Workstation *Mar 2026 – Jun 2026*

AI Developer · University Team Project

- Built the Model Engine connecting text detection, OCR, LLM translation, and inpainting; preserved intermediate outputs to trace image-quality failures.
- Reduced MSE-based pixel damage from **23.15% to 1.50%** with masked compositing and OpenCV post-processing. Cut end-to-end workflow time by approximately **60%** across three manga comparisons.

A.N.S.I., Ajou University *Mar 2024 – Sep 2026*

Member · Algorithm and Programming Club

- Practiced C++ problem solving, joined coding camps and ICPC team training, and competed in SCPC, UCPC, and the ICPC Seoul preliminary contest.
- Placed **21st** in the 2025 shake! finals as an Ajou University representative (held Jan 2026).

Honors and Scholarships

Third Prize (5th overall) *Sep 2026*
5th Inter-University Deep Learning Challenge
Ajou University

Honorable Mention *May 2026*
25th TOPCIT, Ajou University / AI Convergence Education Institute Director's Award

Excellence Award *May 2026*
Ajou Programming Contest (APC), Div. 1
Director's Award, Ajou University SW-Centered University Program

Second Prize (3rd overall) *Oct 2025*
4th Inter-University Deep Learning Challenge
Ajou University (host); Kyungpook National, Kyung Hee, Pusan National, Chonnam National, and Jeonbuk National Universities (co-organizers)

6th Overall / Ajou University First-Place Special Award *Sep 2025*
Centroid Cup / Kookmin, Chung-Ang, Inha, and Ajou Universities

Excellence Award *May 2025*
Ajou Hackathon
Director's Award, Ajou University SW Convergence Education Institute

Two Excellence Awards and One Encouragement Award *2024–2026*
Mogakso, Ajou University
Excellence: Winter 2024, Summer 2025 / Encouragement: Summer 2026

Yulgok Academic Merit Scholarship *2021–2026*
Ajou University / Four semesters / Approximately US$6,900 in total
Spring 2021, Spring 2025, Fall 2025, Spring 2026